

YANSHU ZHANG

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RESEARCH INTERESTS

I build interactive virtual worlds from images and videos, spanning point-based neural rendering, feed-forward 3D reconstruction and novel view synthesis, video-to-4D, and 3D editing and animation.

EDUCATION

Simon Fraser University 09/2021 - Expected 04/2027
Ph.D. in Computer Science (CGPA: 4.26/4.3), supervised by Prof. Ke Li

University of Science and Technology of China 09/2017 - 07/2021
B.S. in Computer Science

PUBLICATIONS

PointWeave: Feed-Forward View Synthesis with Explicit Point Interface
Yanshu Zhang, Shichong Peng, Chirag Vashist, Ke Li Under Submission

RigPAPR: Rig-Based Animation of Static Neural Point Clouds from a Single-View Video
Shichong Peng, **Yanshu Zhang**, Ke Li NeurIPS 2026

The Fault in Our Metrics: Revisiting Generative Model Evaluation with Chamfer Distance
Chirag Vashist, Vaibhav Santhosh, Tristan Engst, **Yanshu Zhang**, Ke Li NeurIPS 2026 E&D

PointGT: Simultaneous Geometry and Texture Editing for Point-Based Representations
Yanshu Zhang, George Shramko, Pratul P. Srinivasan, Ke Li ECCV 2026

P-CORE: Self-Supervised Surface Consistency for Point-Based Neural Editing
Yanshu Zhang, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li ECCV 2026

Tackling Misattribution in 3D Intrinsic Decomposition via Proximity Attention Point Rendering
Alireza Moazeni, Shichong Peng, **Yanshu Zhang**, Chirag Vashist, Ke Li ECCV 2026

PAPR Up-close: Close-up Neural Point Rendering without Holes
Yanshu Zhang, Chirag Vashist, Shichong Peng, Ke Li 3DV 2026

WIMLE: Uncertainty-Aware World Models with IMLE
Mehran Aghabozorgi, Alireza Moazeni, **Yanshu Zhang**, Ke Li ICLR 2026

PROSE: Point Rendering of Sparse-Controlled Edits to Static Scenes
Yanshu Zhang, Shichong Peng, Mehran Aghabozorgi, Alireza Moazeni, Ke Li ECCV 2024 Workshop

PAPR in Motion: Seamless Point-level 3D Scene Interpolation
Shichong Peng, **Yanshu Zhang**, Ke Li CVPR 2024 (**Highlight**)

PAPR: Proximity Attention Point Rendering
Yanshu Zhang*, Shichong Peng*, Alireza Moazeni, Ke Li NeurIPS 2023 (**Spotlight**)

PATENTS

Method and System for Accelerated Operation of Layers Used in a Machine Learning Model and Differentiable Point Rendering Using Proximity Attention

Yanshu Zhang, Yuzhen Mao, Shichong Peng, Alireza Moazeni, Ke Li

CA Patent Application: 3292617 (2024)

EXPERIENCE

National University of Singapore

01/2021 - 05/2021

Research Exchange Programme

TECHNICAL SKILLS

Programming: Python, C++, CUDA (custom kernels)

Frameworks & Tools: PyTorch, multi-GPU distributed training (DDP), SLURM clusters

Research Areas: Neural rendering, NeRF / 3D Gaussian Splatting, point-based rendering, feed-forward 3D reconstruction, novel view synthesis, video-to-4D, 3D editing and animation, generative models

AWARDS

SFU Ph.D. Research Scholarship (2022, 2024)

SFU Graduate Fellowship (2023, 2025)

Silver Award for Outstanding Students of USTC (2019)

Bronze Award for Outstanding Students of USTC (2018)

TEACHING EXPERIENCE

Generative Models, Teaching Assistant, SFU (2025)

Machine Learning, Teaching Assistant, SFU (2022)

ACADEMIC SERVICE

Reviewer for CVPR, ECCV, ICCV, 3DV, SIGGRAPH, SIGGRAPH Asia, ICML, ICLR, NeurIPS, TPAMI